Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



Open Ended Lab

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Date performed: November 11, 2026

MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____
Lab Instructors: Dr. Nasir Ali Shah, Engr. Fasih Ahmad

Code:

```
//-------------------------------------------------------
// File : data_memory.v
// Lab : RISC-V Data Memory (SKETCH - complete it!)
// Name : Syed Sikandar Abbas Roll No: 24I-6553
//-------------------------------------------------------
module data_memory (
	input wire i_clk, // clock
	input wire i_we, // write enable (1 = write)
	input wire [31:0] i_data, // data to be written
	input wire [9:0] i_addr, // word address (0 to 1023)
	output wire [31:0] o_data // data read from memory
);
	// 1) MEMORY ARRAY: 1024 locations, each 32 bits wide
	reg [31:0] mem [0:1023];
	// 2) Note: simulation never shows 'x'
	// 3) WRITE PORT: happens only on the clock edge
	always @(posedge i_clk) begin
		if (i_we == 1) begin
			mem[i_addr] <= i_data;
		end
	end
	// 4) READ PORT: choose ONE style and write why in your report
	// Style A - asynchronous read (data appears at once)
	// assign o_data = mem[i_addr];

		//STYLE B CHOSEN
	// Style B - synchronous read (data appears after a clock edge)
	reg [31:0] r_data;
	always @(posedge i_clk)
		r_data <= mem[i_addr];
	assign o_data = r_data;
endmodule

//-------------------------------------------------------
// File : tb_data_memory.v (SKETCH testbench)
//-------------------------------------------------------
module tb_data_memory();
	// Signals that drive the design
	reg [31:0] i_data;
	wire [31:0] o_data;
	reg [9:0] i_addr;
	reg clk;
	reg i_we;
	// Connect the design (DUT = Design Under Test)
	data_memory dm(
		.i_clk(clk),
		.i_we(i_we),
		.i_data(i_data),
		.i_addr(i_addr),
		.o_data(o_data)
	);
	// Clock: period = 10 ns
	initial begin
		i_addr = 10'b0;
		i_data = 32'b0;
		i_we = 1'b01;
		clk = 1'b1;
```

```
        end
        always #5 clk = ~ clk;
        initial begin
                #10
                i_we = 1;
                i_addr = 10'd1;
                i_data = 32'd25;
                i_we = 1;

                #10
                i_we = 0;
                i_addr = 10'b0;
                i_data = 32'd30;
                i_we = 0;

                #10
                i_we = 0;
                i_addr = 10'b1;
                i_data = 32'd35;
                i_we = 1;

                #10
                i_we = 0;
                i_addr = 10'b1;
                i_data = 32'd40;
                i_we = 1;

                #10
                i_we = 0;
        end
endmodule
```

Output:

